

COMPARATIVE ANALYSIS OF CONVERGENCE TIMES BETWEEN OSPF, EIGRP, IS-IS AND BGP ROUTING PROTOCOLS IN A NETWORK

Eunice Domfeh Asabere ¹, Joseph Kobina Panford ², James Ben Hayfron-Acquah ³

¹ Final Year M Sc. I.T, Dept. Of Computer Science, KNUST, Kumasi, Ghana, easabere@ssnit.org.gh

² Dept. Of Computer Science, KNUST, Kumasi, Ghana, jpanford@yahoo.com

³ Dept. Of Computer Science, KNUST, Kumasi, Ghana, jbha@yahoo.com

Abstract - Convergence time is a key factor in determining performance of routing protocols and routing protocol is one of the significant factor in determining the quality of IP communication. Convergence time is therefore very essential to a network and networks that converge faster are considered to be very reliable. The research was carried out to compare the convergence of four routing protocols namely OSPF, EIGRP, IS-IS and BGP. Network scenarios were created and a simulation was performed using Graphic Network Simulator (GNS3) to measure the convergence times of the protocols separately. Results indicated that EIGRP had the fastest convergence time in both link failure and topology change scenarios. This will help network administrators in their choice of protocols.

Key Words: Convergence time, Protocol, Network, Routing, OSPF, BGP, IS-IS, EIGRP

1. INTRODUCTION

1.1 Background of Study

Data packets traveling through the network typically traverse multiple routers and thus multiple physical links interconnecting them. Whenever there is a link failure or change in topology, routing protocols try to provide an alternative path towards the destination. It is therefore crucial that the routing protocol quickly detects such a link failure or topology change. With the increasing use of networks, any unnecessary loss of connectivity can hardly be tolerated and has to be kept as short as possible. This brings up the issue of convergence time and network is believed to have converged when the routing tables on all routers within the network are complete and correct. Routing protocols play a major role in the delivery of packets from source to destination addresses. In the study, four routing protocols namely Open Shortest Path First (OSPF), Border Gateway Protocol (BGP), Intermediate system to Intermediate system (IS-IS) and Enhanced Interior Gateway Routing Protocol (EIGRP) were compared to determine their convergence time in a given network topology.

1.2 Statement of the Problem

One of the most important characteristics of routing protocols is the convergence time. The convergence time determines how fast the routers adapt their routing tables to topological changes. Among OSPF, EIGRP, IS-IS and BGP, a proof-based advice for selecting the one with the best convergence time is aimed at.

1.3 Research Objectives

The main objectives of this research are:

- To determine the convergence time for OSPF, BGP, IS-IS and EIGRP in a particular network topology.
- To compare the performance of OSPF, BGP, IS-IS and EIGRP.

1.4 Research Questions

The following are the research questions that were posed in order to accomplish the objectives.

- What is the convergence time for OSPF, BGP, IS-IS and EIGRP in a network Topologies?
- Which of the four routing protocols has the fastest convergence time in the topology used.

1.5 Significance of the Study

This study will be significant in providing an in depth understanding of the four routing protocols OSPF, BGP, IS-IS and EIGRP and determining the convergence time of these protocols in a network. The study will also compare the convergence times of these routing protocols and come out with the best one.

2. LITERATURE REVIEW

Convergence can be defined in many ways but in the context of computer networks, a network is said to have converged when all routers in a network have the same topological information about their network they find their selves in. With the help of routing protocols, routers collect topological information [1]. Convergence is a critical property in routing especially dynamic routing. There are about three forms of routing namely static, default and dynamic [2]. A network topology is said to have converged "when routing tables on all routers within the network are complete and correct" [3]. Convergence addresses the manner in which networks recover from problems and network changes. Modern networks anticipate problems by providing alternate, redundant or standby paths.

Convergence time is the time that is required for the routers in a network to learn about routes in a given network. This time is important because it helps administrators of a network to determine in the event that a network downtime occurs due to a failed link between routers or any damage to one router the amount of time it will take for that network to recover and begin to function as a normal network.

Deng et al. [4] performed analysis of RIP and OSPF and EIGRP using OPNET which is a simulator widely used for networking related analysis. In their research, they analysed the performance of these protocols based on their convergence activity, convergence duration and traffic sent

(bytes/sec) to compare the difference in their performance. From their research, they found out that the convergence of EIGRP was faster than the others regardless of the network topology.

Panford *et al.* [5] also analyzed Convergence times between RIP and EIGRP routing protocols in a network using packet tracer which allows network behaviour experimentation and also helps in answering what-if scenarios. In their research, they observed that EIGRP had the fastest convergence time. Their experiments also showed that, regardless of the topology, the convergence time remains the same whether for RIP or EIGRP. Another interesting observation made with EIGRP was that as the number of routers increases, the time for convergence were almost the same.

3. METHODOLOGY

The method for this research was a simulation of scenarios. To help with this simulation, Graphical Network Simulator (GNS3) was employed and the network diagram for the simulation scenarios is illustrated in Fig.1. GNS3 was chosen because it has a user- friendly Graphical User Interface (GUI) and also enables users to configure a network component in a virtual machine that runs the OS same as the original network component.

4. ANALYSIS

The measurements results were placed into three main categories. The first category, based on Fig. 2 consist of measurements of convergence times of protocols with link failure closer to the source of the traffic as shown in Table 1. The second category as derived from Fig 3. is made up of measurements of convergence times of protocols with link failure closer to the destination of the traffic as illustrated in Table 2 and the last category was convergence time measurements under topology change as shown in Table 3. Fig. 4 and Fig. 5 show the network diagram as additional routers are added to the original network diagram.

4.1 Results of Routing protocols with Failure closer to the source of the traffic

Table -1: Convergence time measurement for Protocols with Link Failure closer to the source of the traffic.

Test	OSPF	EIGRP	IS-IS	BGP
1	8.346	6.078	7.799	21.975
2	8.637	5.985	8.340	23.169
3	8.494	5.938	8.368	19.331
4	8.879	5.951	8.185	15.275
5	8.162	5.469	8.590	15.743
6	9.601	6.039	9.095	19.874
7	7.788	5.491	8.532	25.646
8	8.592	5.169	8.042	28.507
9	8.836	7.298	8.051	14.664
10	6.187	6.204	7.441	28.640

4.2 Results of Routing protocols with Failure closer to the destination of the traffic

Table -2: Convergence time measurement for Protocols with Link Failure closer to the source of the traffic.

Test	OSPF	EIGRP	IS-IS	BGP
1	7.993	4.220	8.091	13.553
2	8.490	3.969	8.485	29.517
3	7.797	4.079	8.344	31.220
4	9.078	4.007	8.438	29.993
5	9.005	5.968	8.905	16.394
6	8.673	5.972	8.297	18.859
7	8.938	4.298	7.801	27.502
8	8.841	6.001	8.438	20.646
9	7.735	5.875	8.660	17.641
10	7.704	5.969	7.438	24.204

4.3 Results of Topology Change

Table -3: Convergence time measurement for Protocols with Topology Change

	OSPF	EIGRP	IS-IS	BGP
1	12.138	3.027	8.044	18.227
2	11.117	3.031	12.148	20.258
3	10.913	3.468	11.082	18.229
4	11.530	3.477	9.275	19.541
5	10.026	3.102	9.196	19.205
6	10.993	3.198	10.084	18.714
7	11.362	3.144	9.143	21.095
8	11.122	3.112	8.012	19.521
9	11.212	3.099	8.050	18.008
10	10.410	3.005	7.048	18.035

On the average, it took OSPF network 8.352s to converge, EIGRP network 5.962s to converge, IS-IS network 8.244s to converge and BGP network 21.282s to converge for link failure closer to source of the traffic. For link failure close to the destination of traffic the average convergence times were 8.375s for OSPF, 5.036s for EIGRP, 8.290s for IS-IS and 22.953ms for BGP.

It took an average time of 11.082s for OSPF to converge, 3.166s for EIGRP to converge, 9.208s for IS-IS to converge and 19.083s for BGP to converge for change in topology.

5. CONCLUSION AND RECOMMNDATION

From the simulation results the EIGRP give the best performance. EIGRP generate the least traffic and thus it will consume the least bandwidth, leaving enough bandwidth for transmission of data. EIGRP also has the best performance in the case of topology changes and when there is a broken Ethernet connection.

In conclusion, the simulations confirmed that EIGRP was the best choice for all scenarios implemented as it has a fast convergence, while also efficiently utilizing bandwidth. IS-IS was the second choice as far as convergence time was concerned and then OSFP came next. BGP performed poorly

and is therefore not suitable for large networks. It can therefore be stated based on the results achieved that there is a significant difference in the performance of the protocols as far as convergence time is concerned.

REFERENCES

- [1] Shah A, Waqas J. Rana, "Performance Analysis of RIP and OSPF in Network Using OPNET", International Journal of Computer Science Issues, Issue 6, No 2, November 2013.
- [2] Lammle, Todd, "CCNA Cisco Certified Network Associate study guide, sixth edition". Indianapolis, Ind.: Wiley. (2007).
- [3] Todorovic Ivana, Sepanovic Stevan, "Measurements of convergence time for RIP And EIGRP Protocols", Scripta Scientiarum Naturalium, volume 2, 2011.
- [4] Deng Justice, Wu Siheng, Sun Kenny, "Comparison of RIP, OSPF and EIGRP Routing Protocols based on OPNET" Final project. Spring, 2014.
- [5] Panford, J. K., Riverson K. & Boansi O. K (2015). Comparative analysis of convergence times between rip, and eigrp routing protocols in a network. Research Journal's; Journal of Computer Science., pg 1-5.

APPENDIX

The following are network topologies used in the experiments.

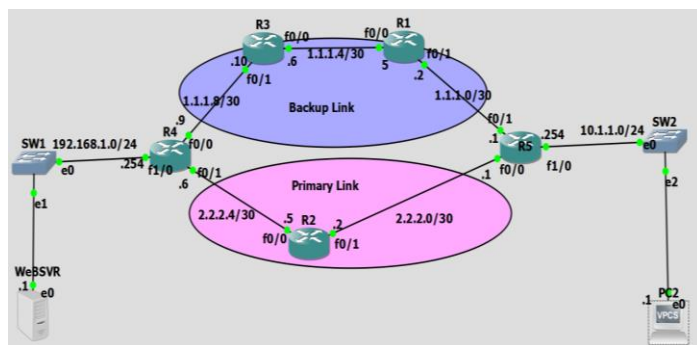


Fig -1: Network diagram for simulation scenarios

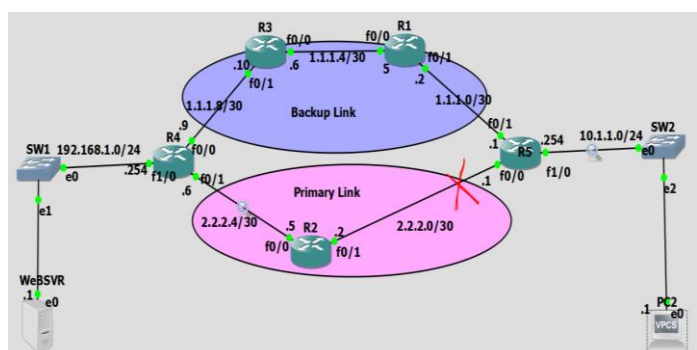


Fig -2: Network diagram for link failure closer to the source of traffic

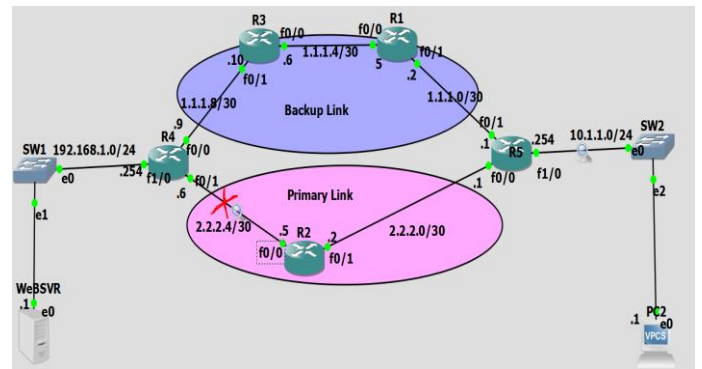


Fig -3: Network diagram for link failure closer to the destination of traffic

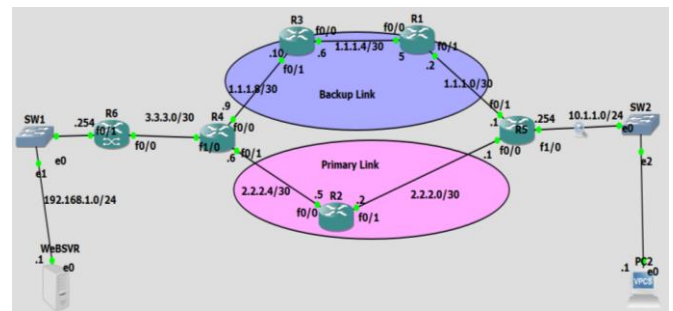


Fig -4: Network diagram with one additional router

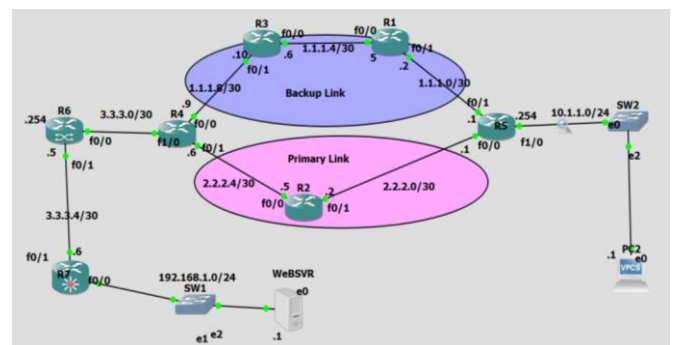


Fig -5: Network Diagram with two additional routers